

Utkrist Gupta

Software Engineer

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Professional Summary

Software Engineer with 3 years of experience in DevOps, and cloud infrastructure. Experienced in AWS, Terraform, Jenkins, Docker, Python, and Linux. Passionate about building highly available, scalable, and automated production systems through Infrastructure as Code, CI/CD, monitoring, and reliability engineering.

Work Experience

Systems Engineer

OCT 2023 – Present

Tata Consultancy Services (TCS)

Bengaluru

- Automated software release and deployment workflows using **Python** and **Selenium**, reducing manual effort by **73%**, improving deployment reliability, and accelerating release cycles.
- Designed, implemented, and maintained end-to-end **Jenkins CI/CD pipelines** for automated build, test, and deployment workflows across Development, QA, and Production environments.
- Provisioned and managed **AWS infrastructure** using **Terraform (Infrastructure as Code)**, enabling consistent, reusable, and version-controlled deployments across multiple environments.
- Managed AWS services including **EC2, IAM, S3, Lambda, and CloudWatch**, ensuring high availability, monitoring infrastructure health, and troubleshooting production issues.
- Developed **AWS Lambda functions** and automation scripts using **Python** and **Bash** to streamline operational activities and reduce manual intervention.
- Built and deployed **containerized applications** using **Docker**, and administered **Linux** servers by performing troubleshooting, log analysis, shell scripting, and deployment automation.
- Collaborated with cross-functional teams using **Git/GitHub** for version control, investigated deployment failures, performed **Root Cause Analysis (RCA)**, and improved system reliability and operational stability.
- Monitored production systems using **CloudWatch**, performed incident response, troubleshooting, and implemented automation to enhance **system reliability**, reduce downtime, and improve deployment performance.

Projects

PageTurn Bookstore — Cloud-Native DevOps Platform | *Java, Spring Boot, Docker, Jenkins, Terraform, AWS* 🐙 [GitHub](#) | 🌐 [Live](#)

- Architected a distributed microservices-based bookstore with 2 Spring Boot REST APIs containerized using Docker multi-stage builds; orchestrated local stack via Docker Compose with bridge networking and persistent volumes.
- Provisioned Jenkins CI/CD server on AWS EC2 using Terraform (IaC); built an automated pipeline (Checkout → Build → Push → Deploy) triggered on every GitHub push.
- Configured Kubernetes Deployments, Services, Health Probes, and Horizontal Pod Autoscaler (HPA) to improve application scalability and availability.
- Monitored containerized applications using CloudWatch and Docker logs to identify and troubleshoot runtime issues.

GenAI-Powered Log Monitoring and Incident Analysis System | *Python, PostgreSQL, Docker, AWS* 🐙 [GitHub](#)

- Designed a centralized log monitoring solution to analyze application logs, detect failures, and improve incident detection and troubleshooting.
- Integrated Gemini API to summarize critical error logs and generate contextual insights, reducing mean time to detection (MTTD) and accelerating incident diagnosis.
- Stored structured logs in PostgreSQL and exposed REST APIs to provide real-time visibility into system health and incident history.

Technical Skills

Language: Python, Bash

DevOps: Jenkins, Jenkins Pipelines, CI/CD Pipelines, Git/GitHub,

Containers: Docker, Kubernetes, Terraform

Cloud: AWS (EC2, S3, VPC, RDS, DynamoDB, Lambda, SNS, SQS, ECR, ALB, IAM, CloudWatch)

Systems: Linux, TCP/IP, HTTP/HTTPS, Load Balancing

Concepts: Microservices, High Availability, Scalability, Infrastructure as Code, Site Reliability, Production Support, Incident Response, Root Cause Analysis

Education

Praveer Singh Institute of Technology

2019 – 2023

Bachelor of Technology in Electronics and Communication Engineering

Certifications

- AWS Certified Solutions Architect - Associate
- Oracle Cloud Infrastructure Certified Generative AI Professional